Leaving for Europe, or Leaving Science?

Reading Parental Leave Policies at Scale to Explain Who Stays in US Academia

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1 Motivation and research questions

Take two women hired as assistant professors in the same year, one at the University of Kentucky and one at the University of Nottingham, two universities of about the same size and the same mix of fields. Both have their first child. In Nottingham the policy gives her 26 weeks off at full pay. In Kentucky she is paid while she recovers, "usually 6–8 weeks", under the faculty rules for temporary disability, and a colleague whose partner gives birth finds no paid leave in those rules at all. US universities also differ among themselves: at CU Boulder either parent gets 18 weeks at full pay. In the US every university writes these rules for itself, while in most of Europe the state writes them and some universities add to them.

I read these documents as culture written down. Besides the number of weeks, they say whom the institution imagines as the parent: a mother who has to recover, any parent who needs time with the child, or a "primary caregiver" who has to declare it. Kentucky's faculty rules treat a birth as a medical event, and Michigan's tenure-clock rule for childbearing is written for "a woman who bears one or more children". The wording is not neutral in its effects either. When top economics departments made tenure-clock extensions gender-neutral, men's tenure rates went up and women's went down (Antecol et al., 2018).

I want to know if these rules change who stays in science. Parenthood explains most of the gender gap in faculty productivity, and women, far more than men, say the leave policy mattered when they chose their job (45.9% against 20.6%, Morgan et al., 2021). But a survey of current faculty only hears from the people who stayed. Following everyone in a census of 245,270 US faculty, Spoon et al. (2023) show that women leave at higher rates at every career age, more often because of workplace climate than because of work-life balance. Leave rules are one part of that climate that a university puts in writing. The census does not show where people went, though. A woman who takes a faculty job in Leuven and a woman who leaves research both disappear from it in the same way, and the destination separates two explanations. If more generous leave abroad pulls women out (the *pull* explanation), their extra departures around the first birth should go to Europe. If a partner's job or the family keeps them in the US, the extra departures should go out of science.

To test this I need the policies themselves, for hundreds of universities, on two continents, for every year a career passes through, and they exist only as text. A leave policy is a set of documents written by HR offices, faculty senates and unions, in different languages and legal traditions, and the number has to be read out of sentences like "120 hours of paid family leave", "a one semester paid leave of absence" for the "primary care giver", "26 weeks' leave on full pay (inclusive of any SMP)" (SMP is the UK's statutory maternity pay) or "Der bezahlte Urlaub beträgt zwei Wochen" ("the paid leave is two weeks"). Morgan et al. (2018) collected and coded the policies of 205 US and Canadian universities in 2018. I propose an NLP pipeline that does this reading at scale and over time: find the documents, keep the passages about leave, extract a fixed set of fields with an LLM that has to quote its evidence, check the extraction against independent coders, and follow the fields across archived versions of the same page. The second half of the project links the extracted policies to where people go when they leave US academia.

RQ1 (policies). How generous are faculty parental leave policies at US universities compared with matched European universities, whom do they address (mothers and fathers, or parents), and how did both change since 2011?

H1 (destination). In the birth window, women's extra departures from US science, relative to men's, go to Europe rather than out of science. *Disconfirmed if* the women-to-men ratio of yearly rates of moving to Europe has a 95% CI that includes or lies below 1, or is smaller than the same ratio for exits.

H2 (policy). These extra departures are smaller where the US university or its state pays at least one semester of leave. *Disconfirmed if* the gender $\times$ birth window $\times$ policy term of the competing-risks model (Section 7) has a 95% CI that does not lie below 0.

H3 (framing). With the weeks held fixed, the framing of a policy changes women's extra departures. The sign is open: gendered wording may tell women the leave is meant for them, while neutral wording also lets fathers use it for research (Antecol et al., 2018). *Disconfirmed if* the gender $\times$ birth window $\times$ framing term has a 95% CI that includes 0.

2 Data

What I used in this pilot. *Policy documents.* I drew 20 US universities at random from the 80 most productive in OpenAlex (Priem et al., 2022), 10 in states with a state paid family leave insurance program and 10 in states without one (these programs started in different years, which Section 7 uses), with CU Boulder fixed in. For each I found a European *twin*: the university closest in its share of output across 26 research fields, its number of works and its citations per work, matched one-to-one with an optimal assignment and at most four twins per country (the twins are in 11 countries, the UK and Switzerland included). The official pages were found with LLM-assisted web search restricted to university and government domains, and then downloaded by a script: faculty handbooks, HR and provost pages, union contracts, and for Europe the national government page when the university page referred to it. In total 92 documents, 551,000 words, in eight languages (English, German, French, Italian, Spanish, Dutch, Swedish, Danish). *The Wayback Machine* I used only in two small ways: to recover pages that blocked automated download, and to count the archived versions of six US policy pages. *Careers.* OpenAlex author records with yearly affiliations:

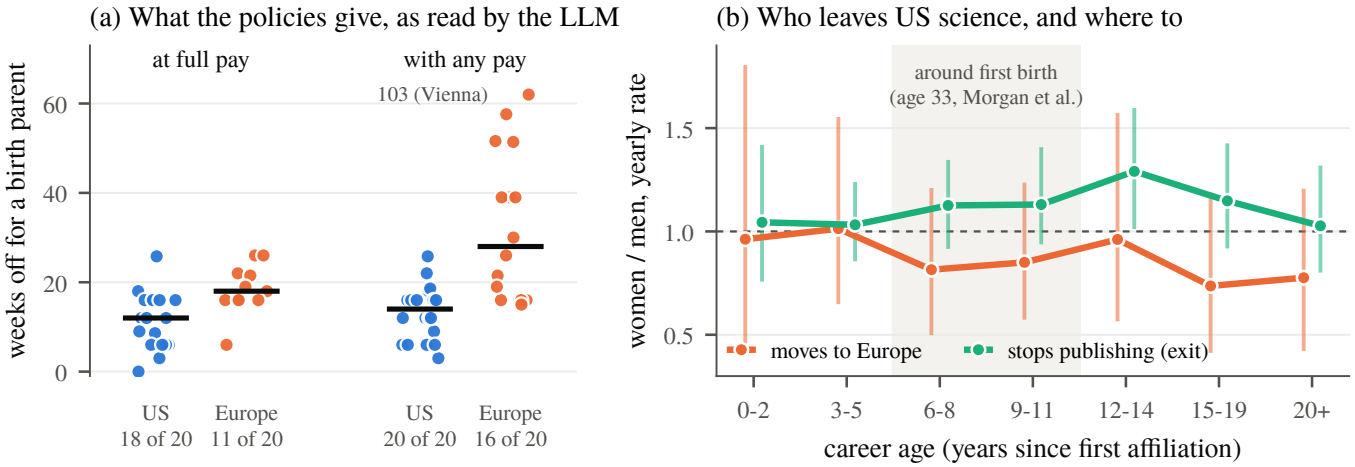


Figure 1: (a) Weeks off for a birth parent at 20 US universities and their 20 European twins, as extracted by the LLM, at full pay and with any pay (bars are medians, "n of 20" counts the texts that state the number). (b) Women-to-men ratio of yearly rates of leaving US science by career age, for moves to Europe and for exits, with 95% CIs from 300 bootstrap resamples of authors. Shaded: the birth window (Section 2).

three random samples of 10,000 authors ever affiliated with a US institution (10 to 1,000 works), and one sample of 10,000 authors ever in the US whose last affiliation is in Europe, the *movers sample*, to have enough moves. Career age is years since an author's first affiliation. Births are not observed, so the *birth window* is career ages 6–11, where a first birth at the faculty average age of 33 (Morgan et al., 2021) falls if careers start in the mid-twenties.

What the full study adds. The AARC faculty census used by Spoon et al. (2023), licensed to a research group at CU, gives who left a US faculty job and when, from 2011, with field, rank and institution, and its hire years give a better career age than first affiliations. OpenAlex then gives where each person went. The 2018 collection of 205 universities (Morgan et al., 2018) is the ground truth for checking the extraction on US policies, and the EPLP dataset (Spitzer et al., 2026) gives the national layer for Europe: maternity, co-parent and parental leave entitlements for 21 countries from 1970 to 2024. The Wayback Machine then becomes a real source: every archived version of every policy page since 2011.

Permissions. The policies are public documents, fetched at a polite rate and not redistributed (the repository has the code to fetch them again). OpenAlex is CC0, and the Wayback Machine is open for research. AARC is licensed, so it would only be used with the agreement of the group that holds it, and linking census records to OpenAlex person by person is human-subjects data that needs IRB review. All results are reported in aggregate.

3 Language features (operationalization)

The NLP part is a pipeline of five steps, and each step is a place where the measurement can fail. (1) *Retrieval*: an LLM agent with web search finds the official policy pages of a university, and a script downloads them, with the Wayback Machine as a fallback. (2) *Passage selection*: union contracts and handbooks run to 50,000–80,000 words, so I cut every document into 120-word windows and keep the windows that mention leave in any of the eight languages (parental, maternity, childbirth, tenure clock, Mutterschutz, Elternzeit, congé maternité, föräldraledighet, and others), with their neighbors. (3) *Extraction*: the LLM reads the passages for one reference person, a tenure-track assistant professor with two years of service and a first child, and fills a codebook of ten fields: weeks off at full pay for the birth parent and for the non-birth parent, weeks with any pay, semesters of teaching relief on top of the leave, the tenure clock (automatic, on request, none), a primary-caregiver clause, who pays, the service needed to qualify, whether the text relies on rules it does not state, and the framing. It converts units (a semester is 16 weeks, a quarter 11, 40 hours one week), and for every field it has to give a verbatim quote of at most 25 words and a confidence. A script checks that each quote really appears in the document. (4) *Framing*: *gendered* if the leave rules use only gendered terms (maternity, paternity, mother, a woman who bears a child), *neutral* if only parent terms (parent, birth parent, caregiver), *mixed* if both. In the full run I would add a flag for birth leave filed under disability or recovery, as in Kentucky. (5) *Versions*: the same extraction run on every archived version of a page, so a change in a field dates a change in the policy.

Why these features. The weeks and the teaching relief are what a woman deciding between two offers would compare, and Morgan et al. (2021) used the same kind of measure, a paid leave of at least 10 weeks, about a semester. The framing is the cultural part of the question: whom the institution expects to do the caring. I compare fields across versions, not pages, because the page around a policy changes much more often than the policy: the NYU handbook page has 31 different page hashes in 33 archived captures. A keyword count cannot do this job, because the weeks hide behind unit conversions, caps ("up to"), conditions ("if the primary caregiver") and references to national law.

4 Feasibility test

Finding the documents: easy in the US, hard in Europe. The matching paired all 20 universities, with one weak pair: Michigan has no European university of its size. I found official rules for all 20 US universities. Nine US pages refused automated download, and the Wayback Machine returned seven of

them. Four of the 20 European twins (Montpellier, Florence, Aix-Marseille, Copenhagen) have no public page on staff leave at all, because the rules sit on intranets behind a staff login, and several more publish only a summary. There, the public text is a sector collective agreement, like the Dutch universities' collective labor agreement (CAO), or the national law itself.

Extracting and checking. Claude Opus 5.5 extracted the codebook for all 40 universities, from national law or the sector agreement where a twin has no public page. On a first pass it reported that Michigan's text was garbled and that Duke's and Case Western's leave sections had been cut off, and both were true: the Wayback copies of Michigan's pages were saved compressed, and my first passage selection kept whole paragraphs and lost the leave sections of the longest handbooks. After I fixed both, every quote passed the script check. As stand-ins for human coders, two models from other companies, GPT-6.1 Terra and Grok 4.7 Fast, coded 10 of the universities blind (five US, five European, Zurich partly in German), from the same texts and with six of the ten fields written as instructions for a human coder. On these 60 cells Claude agrees with GPT on 51 and with Grok on 50, and with at least one of them on 58. All three agree on 9 of 10 birth-parent weeks but only on 5 of 10 framings. Most disagreements are cases the codebook does not decide yet: if a rule the text does not state means zero weeks or a missing value, where mixed framing ends and gendered framing begins, and if the UK rule that a mother must end her maternity leave before her partner can share it is a primary-caregiver clause. I will settle these in the codebook before the full run. Agreement between models is not accuracy, because models can share mistakes, so the full study checks the extraction against human coding (Section 7).

What the policies say. Where the text states it (Figure 1a), a birth parent gets a median of 12 weeks at full pay at the US universities and 18 at their European twins, higher in Europe in 7 of the 9 pairs where both numbers are stated. With any pay, including flat state allowances, it is 14 against 28 weeks. But the European number at full pay can be read from the documents for only 11 of the 20 twins, because most of the other nine point to national law or to a social insurance benefit without stating its level, so the full pipeline has to join the national layer from EPLP. The framing splits by continent: of the 20 US texts 17 are neutral, while of the 20 European texts 18 are gendered (8) or mixed (10). This is the field where the coders agreed least, and for the twins without a public page the text is national law, so this split is the least certain number here. In the full run I would also count gendered and neutral leave terms in the selected passages, as a simple check on the LLM label. For change over time, the archived captures of six US pages range from 3 to 70, and the University of California's policy PDF alone has 9 versions with different content since 2014. I have not yet run the extraction on them.

Careers (H1): no sign of a pull to Europe in the pilot. From the random samples I kept authors whose careers began between 1990 and 2018 and whose first name could be gendered with a name dictionary (gender-guesser, 2016), which is binary and failed on 37.5% of names: 8,943 authors (3,651 women), observed in the US between 2000 and 2020. A *move to Europe* is a last US year followed by at least two years at a European institution, and an *exit* is a last US year with no affiliation anywhere in the five or more years after. In the birth window, the women-to-men ratio for moves to Europe is below 1 (0.81 [0.50, 1.21] at career ages 6–8, 0.85 [0.57, 1.24] at 9–11) and for exits slightly above 1 (1.13 [0.92, 1.35] and 1.13 [0.94, 1.41]), and none of these intervals excludes 1 (Figure 1b). The one exit ratio clearly above 1 is at 12–14 (1.29 [1.01, 1.60]), after the window, close to when tenure is decided. The counts are small (201 moves to Europe by women), names are a weak proxy, and stopping publishing is not the same as leaving academia, so I read this only as a reason for the full test. In the movers sample, 1,699 authors meet the definition of a move to Europe with a career that began in 1990–2018 and a gendered name, and 65% of them began their careers in Europe. Most moves are returns, so the model has to separate returns from first moves to Europe.

5 Relation to existing work

Leave policies have been collected for 205 US and Canadian universities (Morgan et al., 2018) and coded for the top economics departments (Antecol et al., 2018), and for Europe they exist at the level of countries, not universities (Spitzer et al., 2026). Work on faculty careers measures the productivity cost of parenthood (Morgan et al., 2021), attrition and its reasons (Spoon et al., 2023), and the prestige hierarchy that shapes hiring and moves (Wapman et al., 2022), and work on the migration of scholars finds women less internationally mobile than men (Zhao et al., 2023). I found no study that links faculty careers to the leave policies people worked under. On the NLP side, LLM coding matches trained annotators on some tasks and fails on others (Bamman et al., 2024; Than et al., 2025), which is why the extraction is checked by other coders, and Gligorić et al. (2025) show how to draw valid conclusions from LLM labels when they are uncertain, which is how the extracted fields should enter the regressions. What I add is a panel of university leave policies read from text, on both continents and over time, and a link between it and where the people who leave actually go.

6 Methods considered, and rejected

Hand-coding everything, as in 2018: at the pilot's rate of 551,000 words for 40 universities, one snapshot of about 400 US universities and their European matches is around 11 million words, before 15 years of versions, and it would have to be redone every year. *Keyword search for weeks* fails for the reasons in Section 3. *One long prompt with all the policies*: given ten policies in one 30,000-word file, Grok reported four of them as missing, a small case of the long-context failures in Karpinska et al. (2024), so the pipeline reads one university at a time. *Inferring births from paper acknowledgments or from gaps in publishing* is noisy and too private, so I use the survey average instead (Section 2). *Surveys of the people who left* are the best source for their reasons (Spoon et al., 2023), and I keep them for a later stage.

7 Limitations, ethics, next steps

A policy on paper is not its use: 91.8% of eligible mothers but 62.1% of eligible fathers took leave (Morgan et al., 2021), and departments negotiate their own arrangements. OpenAlex affiliations are noisy, and gender from names is binary and fails most on non-European names, which the full study improves with the gender coding used for the census in Spoon et al. (2023). The birth window also overlaps the end of graduate training, and the tenure decision comes soon after, so departures there are high for other reasons. Men at the same career age and field, and the policy contrast in H2, are the comparisons that separate a birth effect from these career stages, and the census hire years give a better career age than first affiliations. The US funding cuts of 2025 and the European programs recruiting US scientists will confound the most recent years, so the main estimates stop before 2025. And the pilot shows associations only.

What I would do next, as the full study. (1) Collect the rules of every PhD-granting US university in AARC and of its closest European match (without the one-to-one rule and the four-per-country cap), every year from 2011, joined to the national layer. (2) Check the extraction against the 2018 collection and against a random 10% of policy versions double-coded by trained student coders or Prolific workers screened with a short quiz, use a panel of LLM coders from different companies to send their disagreements to humans, and carry the remaining error into the regressions (Gligorić et al., 2025). (3) Link the people who leave AARC to their next affiliation in OpenAlex. (4) Fit a competing-risks model of each faculty-year, with five outcomes (stay, move within the US, move to Europe, move elsewhere, exit) and the gender $\times$ birth window $\times$ policy and framing terms of H2 and H3, separating returns home by the country of the first affiliation, with field and the prestige of the US institution as controls (Wapman et al., 2022). For a causal estimate, the state paid leave programs started in different years, from California in 2004 to Colorado in 2024, and the Wayback Machine dates the changes in university policies, so the policy effect can be estimated as a staggered difference-in-differences. (5) From the model, calculate how many exits of women a one-semester floor would have prevented, and what those exits cost a university: a new faculty search, a new start-up package, and the research the person would have produced. (6) Write it up as a short brief for university leaders and state legislators.

Team

Antoni Czolgowski (solo): question, design, data, experiments and writing.

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